

Balaji V

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Summary

Embedded Firmware Engineer with 5+ years of experience in firmware development, IoT systems, and test automation. Hands-on expertise with LPC4078 NXP Microcontroller, Cortex-M, STM32, RA6E1, K24. Skilled in Embedded C, Python, Rust, and FreeRTOS with a strong background in communication protocols (UART, SPI, I2C, MQTT). Adept at linker scripts, system debugging, and mentoring junior engineers. Worked on 8, 16, 32-bit microcontroller and low level driver development like UART, ADC and RTC. Available for immediate joining.

Technical Skills

Languages: Embedded C, Python, Rust

Frameworks/RTOS: FreeRTOS, MQX, TouchGFX, pytest

MCUs: STM32, RA6E1, K24, LPC4078, ATtiny85, ESP32

Tools: Git, IAR EWARM, J-Link, I-Jet, ST-LINK V2, JTAG, SWD, VSC, ADO, AWS, Github CI/CD

LLD/Protocols: UART, I2C, SPI, MQTT, ADC, RTC, RS485, 400 – 900 MHz RF

Areas: Embedded Firmware, IoT, Test Automation, Bootloaders, Linker Scripts

Work Experience

Embedded Firmware Engineer @ Hubbell

Mar 2021 – Sep 2025 (23 Nov 2021 – 17 Sept 2025 on Hubbell-Burndy Payroll)

Engineer I – Embedded Firmware (Apr 2025 – Sep 2025)

- Mentored juniors on coding standards and optimization, boosting bug resolution by 30%.
- Integrated new firmware features into legacy devices, ensuring smooth deployments and improving customer satisfaction without hardware upgrades in field.
- Built LNA feature to handle dual radio signals via SKY65313-21 RF chips on a single antenna.

Engineer II – Embedded Firmware (Apr 2024 – Mar 2025)

- Migrated legacy NV tool to modern IDE, cutting dev time.
- Measured CPU load using DWT_CYCCNT, improving accuracy over timers.
- Developed Python tool to program units without licenses, saving cost/time.
- Tuned dual-band antenna noise band; validated in RF noise + climate chambers.

Associate Embedded Firmware Engineer I (Nov 2021 – Mar 2024)

- Built ADC, RTC, UART drivers for RA6E1/K24 MCUs.
- Generated unit tests catching 90% of bugs pre-release.
- Automated firmware tests with Python (Rearchitect Pytest framework), reducing manual effort by 80%.

- Improved test framework with failure bubble-up and error tracking.
- Documented code/process for cross-time zone teams.
- Presented project at Tech Summit; praised by CTO.
- Supported Git adoption and maintained hardware inventory.

Project Associate (Payroll via Team Lease) (01 Mar 2021 – 22 Nov 2021)

- Built Python apps for DFW process, streamlining tests.
- Integrated PC-Lint, reducing errors and uncovering bugs.
- Consistently delivered clean PRs with minimal review comments.

Junior Embedded Engineer @ XREVOL

17 Dec 2019 – 28 Feb 2021

- Developed cloud-based access control device with RFID, Mobile using LPC4078 and ESP32 using ESP-IDF, LPC XPresso IDE.
- Enhanced firmware for smart door lock systems with STM32 and Silicon Labs BGM13P/BGM111 modules using Simplicity Studio, STMCube and STMCubeMX IDE.

Internship: Embedded Developer @ Greenenvironment

24 May 2019 – 19 Nov 2019

- Designed a real-time turbidity level monitoring system using TCS34725 sensor.

Projects

- Arcade Game Console (ATtiny85) – Low-resource retro game console.
- CubeUNO (Python) – Workflow automation tool; appreciated by Arduino co-founder.
- CDE Your IDE (Python) – Lightweight developer IDE.
- PyomoDoro & StonePaperPi (Python) – GUI-based Pomodoro timer and Human Hand vs Computer game.
- Chess Encryption (Python) – Encrypt files into Chess PGN format with decryption tool.
- Home Automation & Smart Mobile Charger (NodeMCU) – IoT-based home system and safe charging solution.
- Complete IoT Product & BMS for EV – Prototypes using NodeMCU, ThingSpeak & AWS IoT Core.
- RFID Music Player (Arduino UNO) – RFID-controlled music system.
- LCD Display & Register-Level Drivers – Bare-metal programming on Nucleo & Arduino UNO.
- CI/CD for C and Python Projects – GitHub Actions for build, test & code coverage analytics.
- Open Source Contributions – ArduinoFreeRTOS + bug discovery in Brave browser.
- Recognition – Code archived in GitHub Arctic Vault.
- GitHub Blog – Created and maintain free blog site.
- Community Support – Released firmware libraries for multiple dev boards like STM32, HiFive, Photon, 8086, 8085, nrf82840, RaspberryPi, NodeMCU and Arduino.

Education

B.E. Electronics and Communication Engineering
Rajalakshmi Institute of Technology (2015 – 2019)